Michael F. Xu

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Research Interest

My research examines how agentic robots can support human connection and collaboration, and how their behavior can be shaped and revised in relation to people's goals, expectations, and social contexts. I investigate these questions across in-home, public, and laboratory settings, using participatory design, field deployments, and experiments.

Education

Ph.D. **University of Wisconsin–Madison**, Computer Sciences — 2023 – Present
- Advisor: Dr. Bilge Mutlu, Doctoral Minor: Human Ecology

M.Sc. **Georgia Institute of Technology**, Computer Sciences — 2019 – 2021
M.A. **Yale University**, Economics — 2014 – 2016
B.A. **New York University**, Economics, and Mathematics (*summa cum laude)* — 2011 – 2014
B.A. **Tsinghua University**, Economics and Finance — 2010 – 2011

Experience

UW–Madison, Department of Computer Sciences, Graduate Assistant — Madison, WI, 2023 – Present
- Various research and teaching assistant roles within the department.

Van Robotics, Lead Software Architect — Columbia, SC, 2019 – 2023
- Full Stack Engineer (2019 – 2020).
- Design and implement the high-level software architecture of the smart robot tutor ABii; Work with research team to perform data analysis, implement machine learning models, as well as deploying the models to edge devices in real-world settings.

Music X Lab, Research Intern — Remote, 2022 – 2023
- Advised by Dr. Gus G. Xia. Implemented hardware and software solutions for haptic music tutoring systems.

IQVIA, Model Developer — New York, NY, 2017 – 2018
- Develop various Excel models; significant VBA codes used both to enhance user experience, and to compute model outputs (Epidemic/Monte Carlo models).

Publications

Xu, M. F., Zhao, E., Zhang, Y., Michaelis, J. E., Sebo, S., & Mutlu, B. (2026). *Designing Robots for Families: In-Situ Prototyping for Contextual Reminders on Family Routines.* In Proceedings of the 21st ACM/IEEE International Conference on Human-Robot Interaction (HRI) [*23%*] — 2026

Xu, M. F., Cagiltay, B., Hu, Y., Zhu, A., & Mutlu, B. (2026). *Designing Robots to Support Parent-Child Connections: Opportunities Through Robot-Mediated Communication.* In Proceedings of the 25th Interaction Design and Children Conference (IDC) [*23%*]

Xu, M. F., Yang, Q., Kirkorian, H., & Mutlu, B. (2026). *Supporting Family-School Partnerships with Robot-Facilitated Home-Based Activities.* In Proceedings of the 25th Interaction Design and Children Conference (IDC) [*23%*]

Xu, M. F., & Mutlu, B. (2025, March). *Exploring the Use of Robots for Diary Studies.* In 2025 20th ACM/IEEE International Conference on Human-Robot Interaction (HRI) [*25%*] — 2025

He, X., ***Xu, M. F.***, Cagiltay, B., & Mutlu, B. (2025, March). *Developing Robot Prototypes to Explore Robot-Facilitated Family Routines.* In 2025 20th ACM/IEEE International Conference on Human-Robot Interaction (HRI) [LBR]

Xu, M. F., Cagiltay, B., Michaelis, J., Sebo, S., & Mutlu, B. (2024, August). *Robots in family routines: Development of and initial insights from the family-robot routines inventory.* In 2024 33rd IEEE International Conference on Robot and Human Interactive Communication (ROMAN) [*Finalist: Pioneering Research Award*] 2024

Honors & Awards

Special Recognition for Outstanding Reviews, ACM Interaction Design and Children Conference 2026

Computer Sciences Summer Fellowship, University of Wisconsin-Madison 2024

Outreach

Grand Parents University 2025

- Served as the instructor for a two-day session, teaching grandparents and grandchildren how to program robots and interact with large-language models.

UW–Madison Science Expeditions

- Hosted lab tours and stationed exhibition tables, performing multiple interactive robot demonstrations.

Lab Tours for Schools

- Co-hosted many lab tours for classes of students from various grade levels (ranging from elementary to high school). Lab tours include a gentle introduction to the subject of robotics, and interactive demo sessions.

Peer Review Service

Conferences: ACM/IEEE International Conference on Human-Robot Interaction; ACM Conference on Human Factors in Computing Systems; ACM Interaction Design and Children Conference

Journal: ACM Transactions on Human-Robot Interaction

Personal Information

Languages: Mandarin & English

Citizenship: U.S.A.